CONFOEDERATIO,
RESEARCH DIVISION (CRD).

DEMOGRAPHIC AND SETTLEMENT IMPACTS OF THE RUSSIAN INVASION OF UKRAINE.

A Regional Literature Review of Population, Displacement, and War Losses (2022AD - 2026AD).

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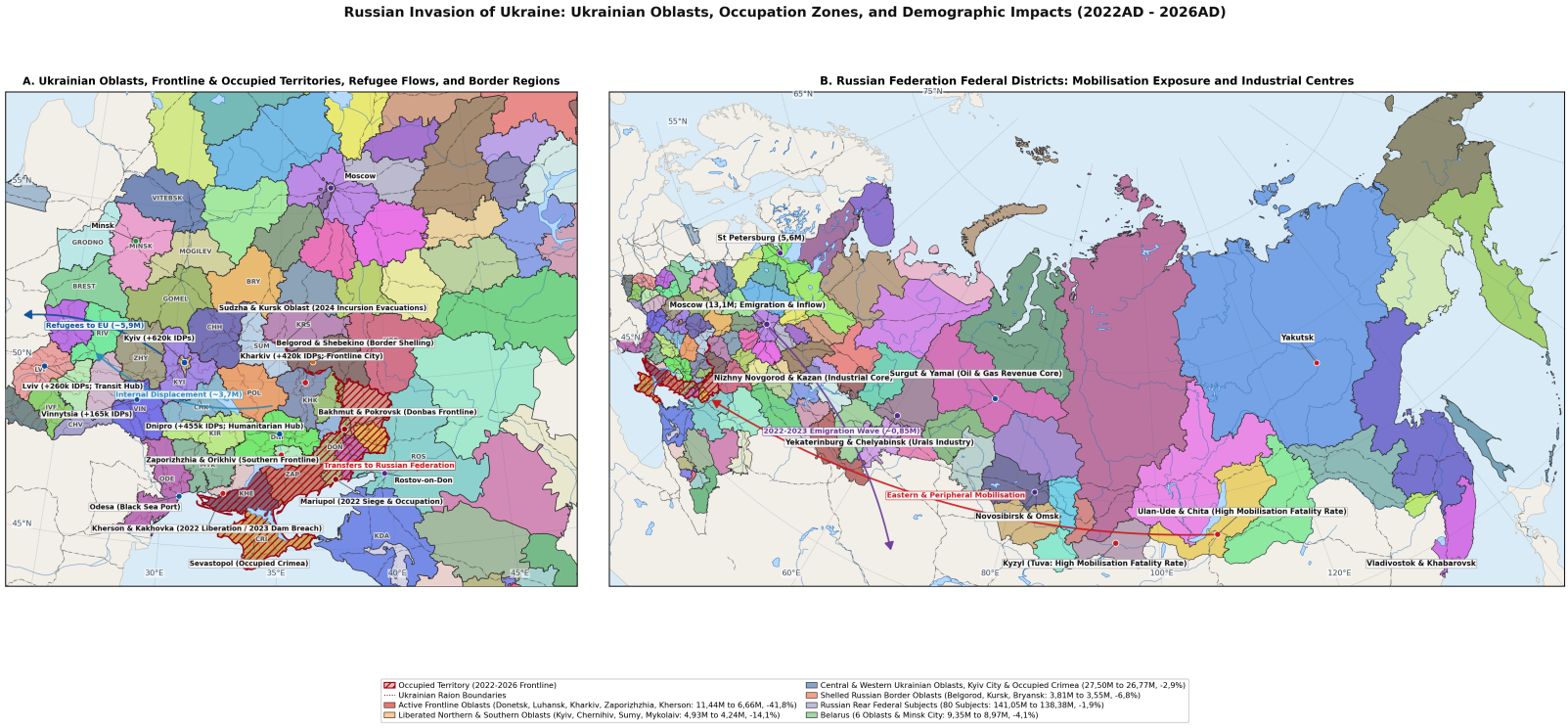


Figure 1. Oblast, occupation-history, and federal-subject divisions across Ukraine, Belarus, and the Russian Federation during the Russo-Ukrainian War (2021AD - 2026AD).

Modern Oblast Architecture & Synthesis

The full-scale military invasion of Ukraine launched by the Russian Federation on 24 February 2022AD triggered the sharpest demographic, settlement, and migratory rupture in Europe since 1945AD [1][2][3][4]. Unlike pre-modern demographic eras shaped by river-basin carrying capacities, every institutional mechanism governing wartime conscription, territorial defence, civilian evacuation, humanitarian registration, fiscal sign-up incentives, and casualty accounting across 2021AD - 2026AD is organised strictly along modern first-level administrative (Admin-1) boundaries - the 27 constitutional Oblasts, Autonomous Republic, and Special-Status Cities of Ukraine, the 7 Oblasts and Capital City of Belarus, and the 83 internationally recognised Federal Subjects (46 Oblasts, 21 Republics, 9 Krais, 4 Autonomous Okrugs, 1 Autonomous Oblast, and 2 Federal Cities) of the Russian Federation.

Ukrainian Regional Military Administrations, Territorial Defence, and Conscription: Under Ukraine's Law *On the Foundations of National Resistance* (Law No. 1702-IX, effective 1 January 2022AD) and Presidential

Decree No. 68/2022 of 24 February 2022AD, every Ukrainian Oblast state administration was converted into an Oblast Military Administration (Oblasna viiskova administratsiia) paired with a dedicated Oblast Territorial Defence Brigade (100th to 128th and 241st Territorial Defence Brigades) and Oblast/Raion Territorial Recruitment and Social Support Centres [1][5]. Mandatory frontline evacuations, curfew regimes, civil-defence fortifications, and eVidnovlennia housing-reconstruction compensation are administered strictly by Oblast.

Oblast-Stratified Internal Displacement (IDP) & Conflict Monitoring: Both the Ukrainian Ministry of Social Policy's Unified IDP Information Database (Yedyna informatsiina baza danykh VPO) and the International Organization for Migration (IOM) Displacement Tracking Matrix (DTM General Population Survey Rounds 1 - 18+) record internally displaced persons (3,67M active IDPs and 4,45M returnees in 2026AD) by modern Oblast of origin and current Oblast of residence, while ACLED conflict events and the DeepState Line of Contact nest directly within Ukraine's modern Oblast and 2020AD-reformed Raion boundaries [4][6][7][8][9].

Russian Federal-Subject Mobilisation Quotas, Governor Contract Bonuses & Border Combat Reserve Units: Inside the Russian Federation, Presidential Decree No. 647 of 21 September 2022AD ("partial mobilisation" of 300 000+ reservists) assigned compulsory call-up quotas to each of the 83 Federal Subjects via regional Military Commissariats (Voennye komissariaty sub'ektov RF) chaired by each Governor [10][11]. Across 2023AD - 2026AD, volunteer contract recruitment (kontraktniki) operated through intense fiscal competition in governor-set regional one-time sign-up bonuses (400k - 800k roubles in poorer oblasts versus 1,9M - 3,20M roubles in Moscow City, Khanty-Mansi AO, Belgorod Oblast, Tatarstan, Bashkortostan, Sverdlovsk, and Leningrad Oblast), while shelled border oblasts raised oblast-specific territorial defence units (in Belgorod, Kursk, and Bryansk oblasts) and Mediazona / BBC News Russian and Meduza / Kobak track combat mortality strictly by Federal Subject [12][13][14].

Accordingly, this synthesis models all 123 exact 10-metre Oblast, Occupation-History, and Federal-Subject polygons (17 959 829 km2, using Ukraine's exact 27 modern Oblasts subdivided strictly where occupation history bifurcated demographic trajectories across Donetsk [government-controlled, post-2022 occupied, and pre-2022 occupied], Luhansk [post-2022 occupied and pre-2022 occupied], Kharkiv [Unoccupied Core, 2022 Liberated & Oskil-Vovchansk East], Zaporizhzhia [Ukrainian North, 2022+ Occupied South], and Kherson [Liberated Right Bank, 2022+ Occupied Left Bank] into 33 Ukrainian divisions alongside 7 Belarusian and 83 Russian Admin-1 units; Figure 1). Across all 123 divisions combined, the de facto resident population contracted by -9 520k (-4,8%), falling from 198 080k (198,08M) in 2021AD to 192 050k in 2022AD, 191 130k in 2023AD, 190 080k in 2024AD, 189 235k in 2025AD, and 188 560k (188,56M) in early 2026AD (Table 1).

Modern Oblast / Federal Subject Tier (Count)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change
UA Frontline Oblast Ukrainian-Held Cores (3)	3 890	2 700	2 785	2 715	2 660	2 635	-32,3%
UA 2022AD Occupied & Liberated Oblast Zones (2)	870	370	335	300	285	275	-68,4%
UA Newly Occupied 2022-2026AD Oblast Zones (4)	3 030	1 270	1 145	1 120	1 110	1 100	-63,7%

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Modern Oblast / Federal Subject Tier (Count)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change
UA Pre-2022 Occupied Donbas & Crimea (4)	6 090	5 240	5 095	5 065	5 025	5 010	-17,7%
UA Feb-Apr 2022AD Defended Border Oblasts (4)	4 930	4 010	4 250	4 250	4 235	4 235	-14,1%
UA Central & Southern IDP Hub Oblasts & Kyiv (7)	13 420	13 090	13 280	13 210	13 140	13 100	-2,4%
UA Western & Northern Rear Oblasts (9)	11 640	12 420	11 850	11 560	11 415	11 305	-2,9%
Belarus Oblasts & Minsk City (7)	9 350	9 230	9 150	9 080	9 020	8 970	-4,1%
RU Shelled Western Border Oblasts (3)	3 810	3 709	3 665	3 603	3 575	3 550	-6,8%
RU Southern Military District Rear Subjects (5)	15 777	15 787	15 757	15 735	15 724	15 711	-0,4%
RU Federal Capitals & Metro Oblasts (4)	29 148	29 001	29 173	29 288	29 315	29 342	+0,7%
RU Hydrocarbon High-Bonus Autonomous Okrugs (2)	2 260	2 275	2 273	2 273	2 274	2 276	+0,7%
RU Defence-Industrial & Core Oblasts/ Krais (39)	63 401	62 650	62 192	61 797	61 456	61 130	-3,6%
RU High-Loss Garrison, Arctic & Pacific Subjects (11)	7 111	6 994	6 934	6 881	6 833	6 786	-4,6%
RU Disproportionate-Loss Ethnic Republics (19)	23 353	23 304	23 246	23 203	23 168	23 135	-0,9%
Ukraine Subtotal (33 Oblast & Occupation Units across 27 Oblasts)	43 870	39 100	38 740	38 220	37 870	37 660	-14,2%
Belarus Subtotal (7 Admin-1 Units)	9 350	9 230	9 150	9 080	9 020	8 970	-4,1%
Russian Federation Subtotal (83 Federal Subjects, 1991AD Borders)	144 860	143 720	143 240	142 780	142 345	141 930	-2,0%

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Modern Oblast / Federal Subject Tier (Count)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change
Total Domain (All 123 Oblast, Occupation & Subject Units)	198 080	192 050	191 130	190 080	189 235	188 560	-4,8%

Table 1. De facto resident population trajectories across the fifteen strategic regional tiers and national subtotals from 2021AD to 2026AD (in thousands).

Proxy, OSINT & Humanitarian Datasets

Because martial law in Ukraine suspended the planned 2023AD All-Ukrainian Census and wartime censorship in the Russian Federation restricted granular Rosstat demographic tables after mid-2022AD [1][13][14][15], reconstructing Oblast-by-Oblast population and settlement series across 2021AD - 2026AD triangulates seven empirical streams.

ACLED Raion & Oblast Conflict Events : Ingests 138 geo-referenced Ukrainian raion polygons aggregating 5 893 high-intensity baseline shelling and combat event clusters and 16 301 geo-located baseline fatalities alongside >180 000 cumulative conflict events recorded by ACLED through 2025AD - 2026AD [8][16]. Fifteen frontline raions in Donetsk, Luhansk, Kherson, Kharkiv, and Kyiv Oblasts account for the overwhelming share of urban kinetic destruction (Table 2 and Figure 2B).

DeepState Line of Contact & Occupied Territory Vectors : Delineates the exact frontline and Russian-occupied share inside Luhansk Oblast (~98% occupied), Donetsk Oblast (~70% occupied), Zaporizhzhia Oblast (~73% occupied), Kherson Oblast (~72% occupied left bank), Kharkiv Oblast (~3% border strip), and AR Crimea / Sevastopol (100% occupied since 2014AD), hatched in crimson on top of intact modern Oblast boundaries in Figure 1 [9][17].

Sentinel-1 SAR Coherence, UNOSAT Damage Assessments & VIIRS Nighttime Lights: Building-level damage classifications from Copernicus Sentinel-1 Synthetic Aperture Radar (SAR) interferometric coherence (Scher & Van Den Hoek), UNOSAT optical assessments, and World Bank rapid damage and needs assessments ($524B reconstruction need across >2{,}5M damaged housing units) are paired with NASA/NOAA VIIRS Day-Night Band radiance series tracking urban depopulation and energy-grid blackouts [18][19][20][21][22][23][24].

Mobile SIM Geolocation, eHealth & Oblast Administrative Registries: For Kyiv-controlled Oblasts, the Ptoukha Institute for Demography and Social Studies (NASU - Ella Libanova, Oleksandr Gladun, Pavlo Shevchuk, Nataliia Levchuk) triangulates active mobile-operator SIM cards (Kyivstar, Vodafone Ukraine, lifecell), eHealth declarations, Oblast IDP subsistence allowances, school enrolments, and municipal water/bread consumption [1][2][5][25][26][27][28].

UNHCR, Eurostat & IOM Oblast Displacement Tracking Matrix (DTM): Cross-border refugee stocks (6,40M global in 2026AD, including ~4,3M EU Temporary Protection beneficiaries) are anchored by UNHCR and Eurostat registers [3][29][30][31][32], while Oblast-by-Oblast IDP (3,67M) and returnee (4,45M) stocks come from IOM General Population Survey Rounds 1 - 18+ [4][6][7][33][34][35][36].

OSINT Probate Registry, Obituary & Excess Mortality Models (Mediazona / BBC / Meduza): Integrates the

Mediazona and BBC News Russian named-obituary database (>105 000 individually verified Russian military dead by 2025AD - 2026AD coded by Federal Subject) with Meduza and Dmitry Kobak's Federal Chamber of Notaries (RND) probate-registry excess mortality model (160 000 - 220 000+ total Russian military dead) [10][11][12][13][14], alongside UALosses (>70 000 named Ukrainian military dead), UN OHCHR civilian casualty monitors (12 600+ verified killed excluding full Mariupol siege tolls of 22 000 - 25 000+), and satellite cemetery measurements [37][38][39][40][41][42][43].

Border-Crossing & Residence-Permit Series on Russian and Belarusian Emigration: Quantifies wartime human-capital outflows (650k - 1,05M peak Russian emigrants, 580k - 680k net retained abroad in 2026AD; 200k - 350k Belarusian outflow) via receiving-state residence permits, tax IDs (IIN), and banking registries across Poland, Lithuania, Germany, Kazakhstan, Armenia, Georgia, Serbia, Turkey, Israel, and the UAE [44][45][46][47][48].

Raion (Parent Oblast)	Oblast ISO	Events	Fatalities	SAR Damage	Primary Frontline Urban Centres & Outcome (2021AD -> 2026AD)
Bakhmutskyi (Donetsk Obl.)	UA-14	1 025	829	88-96%	Bakhmut (71k -> 0,3k), Toretsk (31k -> 1,2k), Soledar (10k -> 0,1k), Chasiv Yar (12k -> 0,3k)
Pokrovskyi (Donetsk Obl.)	UA-14	687	608	62-94%	Avdiivka (31k -> 0,7k), Pokrovsk (60k -> 5,5k), Myrnohrad (46k -> 2,8k), Kurakhove (18k -> 0,9k)
Syevyerodonetskyi (Luhansk Obl.)	UA-09	661	1 493	72-92%	Sievierodonetsk (101k -> 18k), Lysychansk (93k -> 18k), Rubizhne (55k -> 12k), Popasna (19k -> 0,25k)
Kramatorskyi (Donetsk Obl.)	UA-14	455	215	38-85%	Kramatorsk (147k -> 75k), Sloviansk (105k -> 54k), Lyman (20k -> 5,4k), Kostiantynivka (67k -> 12k)
Beryslavskyi (Kherson Obl.)	UA-65	383	943	54-75%	Beryslav (12k -> 1,2k); severe Dnieper right-bank artillery and FPV drone attrition
Izyumskyi (Kharkiv Obl.)	UA-63	314	297	52-68%	Izium (45k -> 23,5k), Balakliia (26k -> 19,8k); liberated in Sept 2022AD counteroffensive
Khersonskyi (Kherson Obl.)	UA-65	262	5 884	41-64%	Kherson City (279k -> 66k), Oleshky (24k -> 8k); June 2023AD Kakhovka dam flood and daily shelling
Volnovaskyi (Donetsk Obl.)	UA-14	238	322	76-95%	Volnovakha (21k -> 10,2k), Vuhledar (14k -> 0,1k); southern Donetsk rail and assault axis
Kharkivskyi (Kharkiv Obl.)	UA-63	215	285	28-78%	Kharkiv City (1 421k -> 1 200k, Saltivka damage and IDP rebound), Chuhuiv (31k -> 24,5k)

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Raion (Parent Oblast)	Oblast ISO	Events	Fatalities	SAR Damage	Primary Frontline Urban Centres & Outcome (2021AD -> 2026AD)
Mariupolskyi (Donetsk Obl.)	UA-14	155	323	74-90%	Mariupol (432k -> 195k incl. Russian garrison); Feb-May 2022AD siege (22k-25k+ civilian dead)
Horlivskyi (Donetsk Obl.)	UA-14	135	11	35-55%	Horlivka (240k -> 155k), Yenakiieve (77k -> 55k); coercive DNR conscription depletion
Buchanskyi (Kyiv Obl.)	UA-32	120	191	26-58%	Bucha (37k -> 38k), Irpin (62k -> 65k); March 2022AD occupation and rapid post-liberation rebuild
Kupyanskyi (Kharkiv Obl.)	UA-63	114	132	55-82%	Kupiansk (27k -> 3,2k), Vovchansk (17k -> 0,15k, May 2024AD razing along northern border)
Donetskyi (Donetsk Obl.)	UA-14	111	627	24-48%	Donetsk City (902k -> 660k), Makiivka (339k -> 250k); frontline water crisis and conscription
Svativskyi (Luhansk Obl.)	UA-09	85	78	42-65%	Svatove (16k -> 7,4k), Kreminna corridor; northern Luhansk logistics and artillery line

Table 2. Top fifteen Ukrainian conflict raions by geo-located combat events, fatalities, satellite building damage, and urban population outcomes (2021AD - 2026AD).

Ukraine: Modern Oblasts and Occupation Zones

Within internationally recognised 1991AD borders, Ukraine's resident population fell by -6 210k (-14,2%) across 2021AD - 2026AD, contracting from 43 870k (43,87M) in early 2021AD to 39 100k in 2022AD, 38 740k in 2023AD, 38 220k in 2024AD, 37 870k in 2025AD, and 37 660k (37,66M) in early 2026AD - comprising ~31,4M in Kyiv-controlled Oblasts and ~6,2M across Russian-occupied Crimea, Luhansk, Donetsk, and southern Zaporizhzhia and Kherson Oblasts [2][3][4]. Table 3 and Figure 2 present the complete Oblast-by-Oblast demographic, Territorial Defence, and IDP trajectory across all 33 Oblast and Occupation-History divisions spanning Ukraine's 27 constitutional Admin-1 Oblasts and Cities.

Five Frontline & Partially Occupied Oblasts (Donetsk UA-14, Luhansk UA-09, Kharkiv UA-63, Zaporizhzhia UA-23, Kherson UA-65; 11 440k -> 6 660k, -41,8%): In Donetsk Oblast (4 060k -> 2 140k, -47,3%; 109th Territorial Defence Brigade), the 2022AD siege of Mariupol (432k -> 195k, 22k - 25k+ civilian dead) [19][40][41], the razing of Bakhmut (71k -> 0,3k), Avdiivka (31k -> 0,7k), Toretsk (31k -> 1,2k), Vuhledar (14k -> 0,1k), and Kurakhove (18k -> 0,9k), and mandatory Donetsk regional military administration evacuations from Pokrovsk (60k -> 5,5k) and Myrnohrad (46k -> 2,8k) halved the oblast's population while

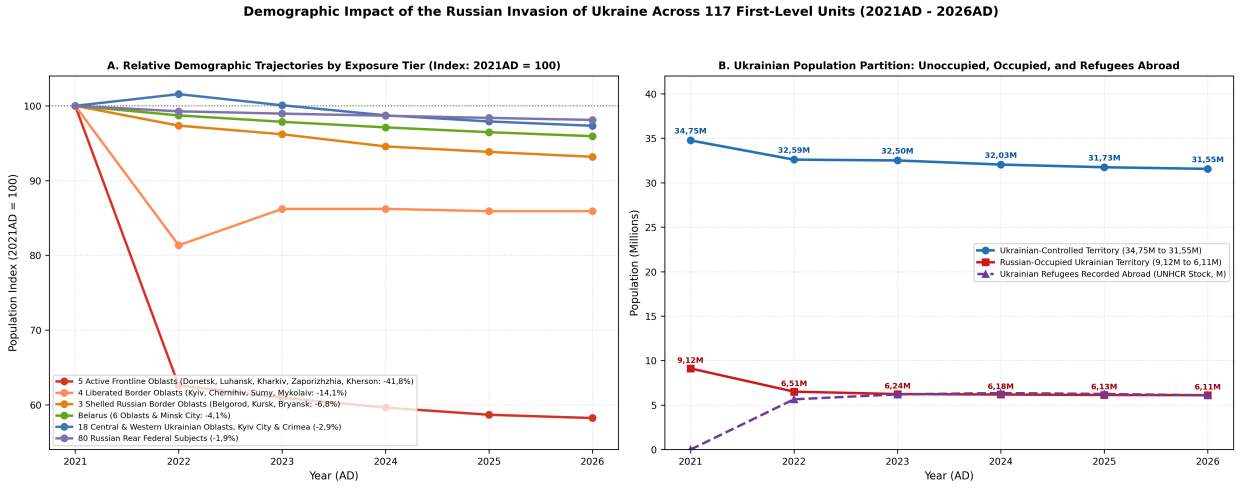


Figure 2. Ukrainian oblast population trajectories comparing 2021AD baselines with 2026AD resident and displaced stocks alongside top frontline conflict raions.

leaving Kramatorsk (147k -> 75k) and Sloviansk (105k -> 54k) as fortified regional administrative hubs [8][9]. Luhansk Oblast (2 100k -> 1 240k, -41,0%; 111th Territorial Defence Brigade) fell almost entirely under Russian occupation after the 2022AD destruction of Popasna (19k -> 0,25k), Sievierodonetsk (101k -> 18k), and Lysychansk (93k -> 18k). Kharkiv Oblast (2 620k -> 1 840k, -29,8%; 113th & 127th Territorial Defence Brigades) defended Kharkiv City (1 421k -> 1 200k, hosting >200k intra-oblast IDPs), liberated Izium (45k -> 23,5k) in September 2022AD, and evacuated Kupiansk (27k -> 3,2k) and razed Vovchansk (17,5k -> 0,15k) during 2024AD - 2025AD border fighting [4][49]. Zaporizhzhia Oblast (1 650k -> 1 060k, -35,8%; 110th Territorial Defence Brigade) remained split along the DeepState front between Ukrainian-held Zaporizhzhia City (710k -> 530k, ~210k IDPs) and Orikhiv (14k -> 1,15k) and Russian-occupied Melitopol (149k -> 110k), Berdiansk (106k -> 74k), and Enerhodar (52k -> 22k). Kherson Oblast (1 010k -> 380k, -62,4%; 124th Territorial Defence Brigade) suffered the steepest relative loss of any Ukrainian Oblast following the November 2022AD right-bank liberation of Kherson City (279k -> 66k), daily cross-Dnieper shelling of Beryslav (12k -> 1,2k), and the 6 June 2023AD Kakhovka dam breach across Oleshky (24k -> 8k) and Nova Kakhovka (45k -> 13k) [50][51].

Four De-Occupied Northern & Southern Border Oblasts (Mykolaiv UA-48, Sumy UA-59, Chernihiv UA-74, Kyiv Oblast UA-32; 4 930k -> 4 235k, -14,1%): Rapid Territorial Defence and regular-army resistance in February - April 2022AD repelled Russian advances on Mykolaiv Oblast (1 090k -> 900k, 123rd Territorial Defence Brigade), Chernihiv Oblast (975k -> 755k, 119th Territorial Defence Brigade), Sumy Oblast (1 035k -> 795k, 117th Territorial Defence Brigade), and Kyiv Oblast (1 830k -> 1 785k, 114th Territorial Defence Brigade). Whereas Kyiv Oblast (~245k IDPs, rapid reconstruction in Bucha 37k -> 38k and Irpin 62k -> 65k) nearly recovered its pre-war baseline, Sumy and Chernihiv Oblasts lost over one-fifth of their residents due to mandatory regional evacuations along the 5 - 15 km shelled Russian border strip in 2023AD - 2026AD [4][8].

Seven Central & Southern IDP Reception Hub Oblasts & Capital (Kyiv City UA-30, Dnipropetrovsk UA-12, Odesa UA-51, Poltava UA-53, Cherkasy UA-71, Kirovohrad UA-35, Vinnytsia UA-05; 13 420k -> 13 100k, -2,4%): Anchored by Kyiv City (2 960k -> 2 965k; 112th & 241st Territorial Defence Brigades, ~375k IDPs) and Dnipropetrovsk Oblast (3 120k -> 3 090k; 108th & 128th Territorial Defence Brigades, Ukraine's No. 1 IDP-hosting oblast with ~455k IDPs in Dnipro 968k -> 980k and Kryvyi Rih 603k -> 585k despite artillery evacuation from Nikopol 105k -> 54k), these seven units absorbed >1,6M IDPs, keeping net 2021AD -

2026AD population change near parity in Poltava Oblast (1 360k -> 1 370k, +0,7%; ~185k IDPs), Cherkasy Oblast (1 170k -> 1 155k, -1,3%), Vinnytsia Oblast (1 520k -> 1 465k, -3,6%), Kirovohrad Oblast (920k -> 875k, -4,9%), and grain-corridor Odesa Oblast (2 370k -> 2 180k, -8,0%) [4][22].

Nine Western Rear & EU Sanctuary Oblasts (Lviv UA-46, Zakarpattia UA-21, Chernivtsi UA-77, Ivano-Frankivsk UA-26, Ternopil UA-61, Khmelnytskyi UA-68, Rivne UA-56, Volyn UA-07, Zhytomyr UA-18; 11 640k -> 11 305k, -2,9%) & Crimea (UA-43, UA-40; 2 440k -> 2 360k, -3,3%): Zakarpattia Oblast (1 250k -> 1 290k, +3,2%; 101st Territorial Defence Brigade), Lviv Oblast (2 490k -> 2 520k, +1,2%; 103rd & 125th Territorial Defence Brigades, ~260k IDPs), and Chernivtsi Oblast (890k -> 900k, +1,1%; 107th Territorial Defence Brigade) recorded net wartime demographic expansion via IDP settlement and enterprise relocation, whereas northern forest-belt Volyn (-7,7%), Rivne (-8,6%), and Zhytomyr (-10,9%) experienced cross-border refugee departures alongside northern border fortification duty [3][4]. Meanwhile, AR Crimea (1 930k -> 1 875k) and Sevastopol City (510k -> 485k) contracted under Black Sea Fleet strikes and Crimean Tatar draft flight.

Ukrainian Oblast & Occupation Zone (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Territorial Defence, Administration & Displacement Profile
Donetsk Oblast - Ukrainian-Controlled GCA (UA-14-UKR)	760	330	290	260	235	220	-71,1%	109th TrO / Donetsk OVA; Kramatorsk-Sloviansk fortified hub & Pokrovsk-Myrnohrad evacuations
Donetsk Oblast - Occupied 2022-2026AD (UA-14-OCC22)	1 080	310	280	285	290	290	-73,1%	2022AD Mariupol siege (22k-25k+ dead), razed Bakhmut, Avdiivka, Vuhledar & Kurakhove
Donetsk Oblast - Pre-2022 Occupied ORDLO (UA-14-ORDLO)	2 220	1 730	1 670	1 655	1 635	1 630	-26,6%	Occupied since 2014AD (Donetsk, Makiivka, Horlivka); Feb 2022AD total male conscription & water crisis

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Ukrainian Oblast & Occupation Zone (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Territorial Defence, Administration & Displacement Profile
Luhansk Oblast - Occupied 2022-2026AD (UA-09-OCC22)	670	240	225	220	220	220	-67,2%	111th TrO evacuated; 2022AD destruction of Sievierodonetsk, Lysychansk, Rubizhne & Popasna
Luhansk Oblast - Pre-2022 Occupied ORDLO (UA-09-ORDLO)	1 430	1 090	1 035	1 030	1 020	1 020	-28,7%	Occupied since 2014AD (Luhansk, Alchevsk, Kadiivka); coercive 2022AD LNR male mobilisation
Kharkiv Oblast - Unoccupied Core & South (UA-63-UKR)	2 210	1 600	1 745	1 730	1 710	1 705	-22,9%	113th & 127th TrO / Kharkiv OVA; Kharkiv City (420k Oblast IDPs), Chuhuiv & Lozova rail hub
Kharkiv Oblast - 2022AD Liberated East & Oskil Front (UA-63-DEOCC)	410	180	175	150	140	135	-67,1%	Occupied Feb-Sept 2022AD (Izium, Balakliia, Kupiansk); 2024-2026AD Vovchansk & Oskil evacuations
Zaporizhzhia Oblast - Ukrainian-Controlled North (UA-23-UKR)	920	770	750	725	715	710	-22,8%	110th TrO / Zaporizhzhia OVA; Zaporizhzhia City (210k IDPs) & Orikhiv-Huliaipole defensive line

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Ukrainian Oblast & Occupation Zone (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Territorial Defence, Administration & Displacement Profile
Zaporizhzhia Oblast - Occupied South 2022-2026AD (UA-23-OCC22)	730	410	380	365	355	350	-52,1%	Occupied since Mar 2022AD (Melitopol, Berdiansk, Enerhodar ZNPP, Tokmak); mass IDP exodus north
Kherson Oblast - Liberated Right Bank (Nov 2022AD) (UA-65-LIB22)	460	190	160	150	145	140	-69,6%	124th TrO; Occupied Mar-Nov 2022AD, liberated Nov 2022AD; Kherson City & Beryslav Dnieper shelling
Kherson Oblast - Occupied Left Bank 2022-2026AD (UA-65-OCC22)	550	310	260	250	245	240	-56,4%	Occupied since Feb 2022AD (Nova Kakhovka, Oleshky, Skadovsk, Henichesk); June 2023AD Kakhovka flood
Mykolaiv Oblast (UA-48)	1 090	790	860	880	890	900	-17,4%	123rd TrO / Mykolaiv OVA; 2022 coastal defence, potable water crisis & 125k returnees/ IDPs
Sumy Oblast (UA-59)	1 035	840	860	830	805	795	-23,2%	117th TrO / Sumy OVA; April 2022 liberation & 2023-2026 10 km border shelling evacuations

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Ukrainian Oblast & Occupation Zone (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Territorial Defence, Administration & Displacement Profile
Chernihiv Oblast (UA-74)	975	780	800	780	765	755	-22,6%	119th TrO / Chernihiv OVA; Feb-April 2022 siege & northern border FPV/artillery depopulation
Kyiv Oblast (UA-32)	1 830	1 600	1 730	1 760	1 775	1 785	-2,5%	114th TrO / Kyiv OVA; Bucha-Irpin-Hostomel liberation, rapid rebuild & 245k registered IDPs
Kyiv City (UA-30)	2 960	2 550	2 850	2 920	2 945	2 965	+0,2%	112th & 241st TrO / KMVA; air-defended capital rebound & 375k registered IDPs
Dnipropetrovsk Oblast (UA-12)	3 120	3 150	3 180	3 140	3 110	3 090	-1,0%	108th & 128th TrO / Dnipropetrovsk OVA; No. 1 IDP oblast (455k IDPs), medical/ logistics hub & shelled Nikopol
Odesa Oblast (UA-51)	2 370	2 150	2 180	2 180	2 180	2 180	-8,0%	122nd & 126th TrO / Odesa OVA; 215k IDPs, Black Sea & Izmail Danube maritime grain corridor
Poltava Oblast (UA-53)	1 360	1 480	1 430	1 400	1 380	1 370	+0,7%	116th TrO / Poltava OVA; 185k IDPs from Kharkiv and Donbas (No. 1 per-capita central IDP hub)

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Ukrainian Oblast & Occupation Zone (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Territorial Defence, Administration & Displacement Profile
Cherkasy Oblast (UA-71)	1 170	1 230	1 200	1 180	1 165	1 155	-1,3%	118th TrO / Cherkasy OVA; 135k IDPs in Cherkasy and Uman central Dnieper corridor
Kirovohrad Oblast (UA-35)	920	950	920	900	885	875	-4,9%	121st TrO / Kirovohrad OVA; 85k IDPs in Kropyvnytskyi and Oleksandriia
Vinnytsia Oblast (UA-05)	1 520	1 580	1 520	1 490	1 475	1 465	-3,6%	120th TrO / Vinnytsia OVA; 165k IDPs, enterprise relocation & Air Force HQ rear
Zhytomyr Oblast (UA-18)	1 190	1 110	1 100	1 080	1 070	1 060	-10,9%	115th TrO / Zhytomyr OVA; northern Belarusian border fortifications & 60k IDPs
Khmelnytskyi Oblast (UA-68)	1 240	1 290	1 230	1 200	1 185	1 175	-5,2%	106th TrO / Khmelnytskyi OVA; 125k IDPs, nuclear power hub & logistics corridor
Rivne Oblast (UA-56)	1 160	1 130	1 100	1 080	1 070	1 060	-8,6%	104th TrO / Rivne OVA; northern border defence, Rivne NPP & 52k IDPs
Volyn Oblast (UA-07)	1 040	1 050	1 000	980	970	960	-7,7%	100th TrO (Mech.) Bde / Volyn OVA; Polish Yahodyn transit & Belarusian border shield

(Continued from previous page)

Ukrainian Oblast & Occupation Zone (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Territorial Defence, Administration & Displacement Profile
Lviv Oblast (UA-46)	2 490	2 820	2 650	2 580	2 545	2 520	+1,2%	103rd & 125th TrO / Lviv OVA; 260k IDPs, primary EU humanitarian/ rail hub & IT relocation
Ivano-Frankivsk Oblast (UA-26)	1 350	1 490	1 410	1 370	1 355	1 340	-0,7%	102nd TrO / Ivano-Frankivsk OVA; 120k IDPs & relocated industrial enterprises
Ternopil Oblast (UA-61)	1 030	1 110	1 060	1 030	1 010	1 000	-2,9%	105th TrO / Ternopil OVA; 68k IDPs vs cross-border labour/refugee outflow
Zakarpattia Oblast (UA-21)	1 250	1 420	1 350	1 315	1 300	1 290	+3,2%	101st TrO / Zakarpattia OVA; +3,2% net growth, 145k IDPs & Carpathian factory relocations
Chernivtsi Oblast (UA-77)	890	1 000	950	925	910	900	+1,1%	107th TrO / Chernivtsi OVA; +1,1% net growth, 80k IDPs & Romanian Siret transit
Autonomous Republic of Crimea (UA-43)	1 930	1 915	1 895	1 890	1 885	1 875	-2,8%	Occupied since 2014AD; Crimean Tatar mobilisation flight & North Crimean Canal loss

(Continued from previous page)

Ukrainian Oblast & Occupation Zone (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Territorial Defence, Administration & Displacement Profile
Sevastopol City (UA-40)	510	505	495	490	485	485	-4,9%	Occupied Black Sea Fleet HQ; missile/USV strikes & partial fleet relocation to Novorossiysk
Ukraine Total (All 33 Oblast & Occupation Units / 27 Oblasts)	43 870	39 100	38 740	38 220	37 870	37 660	-14,2%	UNHCR 6,4M refugees abroad; IOM 3,67M IDPs & 4,45M returnees

Table 3. De facto resident population trajectories across all thirty-three Ukrainian oblast and occupation-history divisions from 2021AD to 2026AD (in thousands).

Displacement / Demographic Category	Peak 2022-2023AD	Early 2026AD Stock	Primary Empirical Source & Oblast Geographic Distribution
Ukrainian Refugees Abroad (Total Global)	6 800k	6 400k	UNHCR Operational Data Portal (4,3M EU TPD: Germany 1,20M, Poland 0,98M, Czechia 0,38M; UK/US/Canada 0,9M; Russia/Belarus 1,2M)
Internally Displaced Persons (IDPs by Oblast)	5 900k	3 670k	IOM DTM Oblast Registers (Dnipropetrovsk 455k, Kharkiv 420k, Kyiv City 375k, Lviv 260k, Kyiv Obl. 245k, Odesa 215k, Poltava 185k)
Returned Displaced Ukrainians (IDP + Foreign)	4 100k	4 450k	IOM Returns Report (78% from internal displacement, 22% from abroad; concentrated in Kyiv City/Obl. 1,1M, Kharkiv 680k, Mykolaiv, Chernihiv)
Russian Border Oblast Evacuees (RU-BEL / RU-KRS)	65k	145k	Belgorod Oblast (Shebekino, Graivoron 2023-2025AD) and Kursk Oblast (Aug 2024AD Sudzha incursion: 130k-150k civilians evacuated)
Russian Wartime Emigration (Brain Drain)	820k - 1 050k	580k - 680k	Receiving-state registries: EU (240k), Israel (135k), Kazakhstan (130k), Turkey (102k), Serbia (92k), Armenia (85k), UAE (70k), Georgia (68k)

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Displacement / Demographic Category	Peak 2022-2023AD	Early 2026AD Stock	Primary Empirical Source & Oblast Geographic Distribution
Belarusian Political & Wartime Outflow (7 Units)	310k	290k	Eurostat Polish (165k) and Lithuanian (62k) residence permits and IT sector relocation from Minsk, Brest, and Grodno Oblasts

Table 4. Wartime external refugee, internal displacement, returnee, border evacuation, and emigration stocks across Ukraine, Russia, and Belarus (2022AD - 2026AD).

Belarus & Shelled Russian Border Oblasts

Disaggregating Belarus into its seven modern Admin-1 divisions (6 Oblasts plus Minsk City; Table 5) and isolating Russia's three shelled western border Oblasts (Belgorod RU-BEL, Kursk RU-KRS, Bryansk RU-BRY) captures sharp frontier asymmetries obscured by national totals.

All Seven Belarusian Oblasts & Minsk City (9 350k -> 8 970k, -4,1%): In February 2022AD, Russian forces launched the northern Kyiv and Chernihiv offensive from staging bases in Gomel Oblast (1 330k -> 1 250k, -6,0%; Gomel, Mozyr, Zyabrovka) and airfields in Brest Oblast (1 340k -> 1 285k, -4,1%; Luninets, Baranovichi) [9]. Across 2021AD - 2026AD, Belarus lost -380k residents through a combination of deep natural decrease (-55k to -65k annually at TFR 1,16) in Vitebsk Oblast (-6,2%) and Mogilev Oblast (-6,2%) and the outflow of 200k - 350k IT professionals, physicians, and skilled workers from Minsk City (2 010k -> 1 980k) and Grodno Oblast (1 020k -> 965k, -5,4%) to Poland and Lithuania [29][48][52].

Russian Shelled Border Oblasts (Belgorod, Kursk, Bryansk; 3 810k -> 3 550k, -6,8%): In 2023AD - 2025AD, artillery and drone bombardment along the Kharkiv-Belgorod frontier forced civilian evacuations from Shebekino (39,6k -> 24,0k) and Graivoron and reduced Belgorod Oblast (1 540k -> 1 445k, -6,2%), where regional authorities raised Belgorod border reserve territorial defence battalions and hiked one-time contract bonuses to 3,0M roubles [8][11]. In August 2024AD, the Ukrainian cross-border offensive into Kursk Oblast (1 082k -> 985k, -9,0%) captured Sudzha (5,1k -> 0,6k) and displaced 130 000 - 150 000 residents toward Kursk City (440k -> 425k), Zheleznogorsk, and Oryol, while Bryansk Oblast (1 188k -> 1 120k, -5,7%; Bryansk border reserve) contracted under border shelling and natural decrease [9][53].

Belarusian Oblast / Capital (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Wartime Staging, Sanctions & Migration Profile
Minsk City (BY-HM)	2 010	1 995	1 990	1 988	1 984	1 980	-1,5%	Capital metropolis; IT/ medical brain drain (EPAM, Wargaming exit to Poland/ Lithuania) offset by internal rural inflow

(Continued from previous page)

Belarusian Oblast / Capital (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Wartime Staging, Sanctions & Migration Profile
Minsk Oblast (BY-MI)	1 470	1 460	1 455	1 452	1 448	1 445	-1,7%	Suburban capital ring & Salihorsk potash sanctions shock
Brest Oblast (BY-BR)	1 340	1 325	1 312	1 302	1 293	1 285	-4,1%	Polish/Volyn border oblast; Luninets & Baranovichi airbase staging & cross-border emigration
Grodno Oblast (BY-HR)	1 020	1 005	993	983	974	965	-5,4%	Polish-Lithuanian borderland; heavy working-age emigration via Karta Polaka & EU permits
Vitebsk Oblast (BY-VI)	1 210	1 185	1 170	1 155	1 141	1 135	-6,2%	Northern industrial oblast; severe natural decrease (TFR 1,14) & out-migration
Gomel Oblast (BY-HO)	1 330	1 308	1 290	1 273	1 262	1 250	-6,0%	Primary Feb 2022 Kyiv/ Chernihiv invasion staging zone (Gomel, Mozyr, Zyabrovka)
Mogilev Oblast (BY-MA)	970	952	940	927	918	910	-6,2%	Eastern oblast (Mogilev, Bobruisk); military training grounds & demographic ageing

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Belarusian Oblast / Capital (ISO)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Wartime Staging, Sanctions & Migration Profile
Belarus Total (All 7 Consti-tutional Admin-1 Units)	9 350	9 230	9 150	9 080	9 020	8 970	-4,1%	Natural decrease (-59k/ yr) & 290k net working-age outflow to Poland/ Lithuania

Table 5. Resident population trajectories across all seven Belarusian oblasts and Minsk City from 2021AD to 2026AD (in thousands).

Russian Federal Subjects: Mobilisation & Mortality

Across all 83 internationally recognised Federal Subjects of the Russian Federation (1991AD borders, excluding occupied Ukrainian territories), total resident population declined by -2 930k (-2,0%) from 144 860k in 2021AD to 141 930k in 2026AD (Tables 6 and 8; Figure 3). Because mobilisation quotas, governor sign-up bonuses, defence-industrial exemptions, and garrison deployments are organised strictly by Federal Subject, wartime demographic impacts varied starkly across five subject groups.

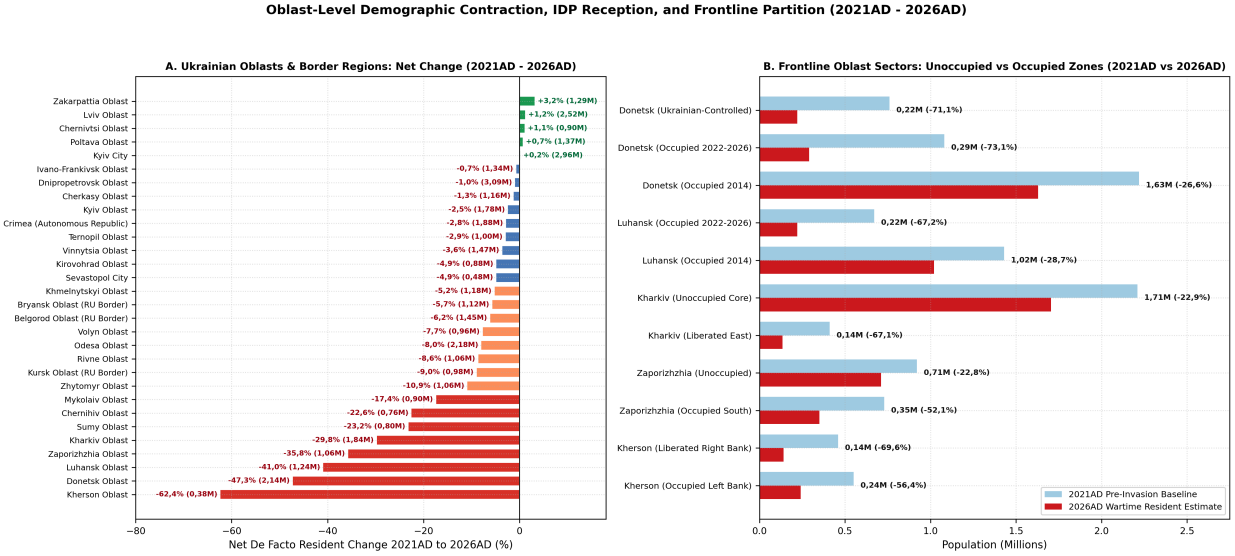


Figure 3. Confirmed military fatalities per 10 000 working-age males across Russian federal subjects and wartime emigration by destination (2022AD - 2026AD).

Federal Capitals vs Brain-Drain Outflow (Moscow City RU-MOS, Moscow Oblast RU-MOW, Saint Petersburg RU-SPE, Leningrad Oblast RU-LEN): To insulate metropolitan populations from political

backlash, authorities kept compulsory mobilisation minimal in Moscow City (3,2 confirmed deaths per 10 000 males; 13 010k -> 12 880k in 2022AD before rebounding to 13 120k in 2026AD) and Saint Petersburg (6,8 per 10 000; 5 602k -> 5 565k), substituting 1,9M - 2,5M+ rouble contract bonuses while experiencing >65% of Russia's 650k - 1,05M skilled wartime emigration [10][44][45].

Disproportionate-Loss Ethnic Republics (19 Republics): Econometric analysis of Mediazona / BBC obituaries and Meduza / Kobak probate excess male mortality confirms extreme ethnic and rural socio-economic stratification [10][12][13][14]. Per-capita confirmed mortality peaked in Tuva (RU-TY, 122,0 per 10 000 males, 38,1x Moscow; 55th Mountain Brigade in Kyzyl), Buryatia (RU-BU, 96,0 per 10 000, 30,0x Moscow; 36th CAA in Ulan-Ude and Kyakhta), and the Altai Republic (RU-AL, 82,0 per 10 000, 25,6x Moscow). In the Volga-Ural and Caucasus belt, Bashkortostan (RU-BA, 4 091k -> 4 015k) suffered the highest absolute military death toll of any Russian Federal Subject (>4 850 named dead; 8 700 - 10 500+ total dead; 44,5 per 10 000), alongside heavy losses in North Ossetia-Alania (RU-SE, 56,0 per 10 000), Kalmykia (RU-KL, 46,0), Mari El (RU-ME, 43,8), Chuvashia (RU-CU, 42,5), Udmurtia (RU-UD, 36,2), Sakha (Yakutia) (RU-SA, 35,0), Dagestan (RU-DA, 29,5, where TFR ~1,95 still drove net population growth 3 180k -> 3 295k), and Tatarstan (RU-TA, 26,5, where 2,0M+ rouble bonuses and Alga/Timer battalion losses coincided with Kazan aviation and Alabuga Shahed drone expansion) [11][12].

High-Loss Airborne, Northern & Pacific Garrison Subjects (11 Subjects): Permanent brigade home stations in Zabaykalsky Krai (RU-ZAB, 74,0 per 10 000; 29th CAA in Chita and Borzya), Sakhalin Oblast (RU-SAK, 62,5; 39th Brigade), Chukotka AO (RU-CHU, 58,0), Magadan Oblast (RU-MAG, 54,0), Pskov Oblast (RU-PSK, 49,5; 76th VDV), Kamchatka Krai (RU-KAM, 48,5; 40th Naval Infantry), Murmansk Oblast (RU-MUR, 46,2; 200th Arctic Brigade), and Primorsky Krai (RU-PRI, 41,5; Vladivostok 155th Naval Infantry and Ussuriysk 5th CAA) suffered severe combat attrition compounded by Far Eastern and Arctic out-migration [12][53].

Hydrocarbon Fiscal-Bonus Autonomous Okrugs (Khanty-Mansi AO RU-KHM, Yamalo-Nenets AO RU-YAN): Offering record 2,75M - 3,20M rouble regional sign-up bonuses (2024AD - 2025AD) to fulfil federal recruitment quotas via out-of-region volunteers while shielding Surgut, Nizhnevartovsk, and Novy Urengoy oil-and-gas shift crews, both okrugs sustained slight net demographic growth (2 255k -> 2 290k, +1,6%) [11].

Defence-Industrial & Southern Rear Oblasts/Krais (44 Subjects): Round-the-clock three-shift munitions, tank, and aircraft production in Sverdlovsk Oblast (RU-SVE, Nizhny Tagil Uralvagonzavod), Chelyabinsk Oblast (RU-CHE), Perm Krai (RU-PER), Tula Oblast (RU-TUL), Nizhny Novgorod Oblast (RU-NIZ), Samara Oblast (RU-SAM), Novosibirsk Oblast (RU-NVS), Omsk Oblast (RU-OMS), Krasnoyarsk Krai (RU-KYA), Irkutsk Oblast (RU-IRK), and Khabarovsk Krai (RU-KHA) stabilised core industrial cities even as rural districts and penal/contract cohorts absorbed heavy casualties (30 - 43 confirmed dead per 10 000 males), while Southern Military District rear hubs in Krasnodar Krai (RU-KDA, 5 838k -> 5 935k) and Rostov Oblast (RU-ROS, 4 201k -> 4 145k) anchored military logistics [12][53].

Russian Federal Subject (ISO)	Named Dead	Est. Total Dead	Deaths / 10k	vs Moscow	Federal-Subject Socioeconomic & Garrison Driver
Tuva Republic (RU-TY)	1 450	2 600 - 3 200	122,0	38,1x	Acute rural poverty; Kyzyl 55th Mountain Brigade & high contract enlistment
Buryatia Republic (RU-BU)	3 120	5 600 - 6 800	96,0	30,0x	Ulan-Ude & Kyakhta 36th CAA / 11th VDV attrition & cross-border flight to Mongolia

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Russian Federal Subject (ISO)	Named Dead	Est. Total Dead	Deaths / 10k	vs Moscow	Federal-Subject Socioeconomic & Garrison Driver
Altai Republic (RU-AL)	920	1 650 - 2 000	82,0	25,6x	Rural South Siberian indigenous republic; household debt & contract bonuses
Zabaykalsky Krai (RU-ZAB)	2 640	4 800 - 5 800	74,0	23,1x	Chita & Borzya 29th CAA (36th Bde) & depressed mono-industrial mining towns
Sakhalin Oblast (RU-SAK)	1 080	1 900 - 2 400	62,5	19,5x	Yuzhno-Sakhalinsk 39th Motor Rifle Brigade & Kuril garrison deployments
North Ossetia-Alania (RU-SE)	1 180	2 100 - 2 600	56,0	17,5x	Vladikavkaz 19th Motor Rifle Division & North Caucasus volunteer battalions
Pskov Oblast (RU-PSK)	1 190	2 100 - 2 600	49,5	15,5x	Pskov 76th Guards Air Assault Division (VDV) & 2nd Spetsnaz Brigade losses
Murmansk Oblast (RU-MUR)	960	1 700 - 2 100	46,2	14,4x	Pechenga 200th Arctic Brigade & Northern Fleet 61st Naval Infantry losses
Bashkortostan Republic (RU-BA)	4 850	8 700 - 10 500	44,5	13,9x	No. 1 in absolute Russian war dead; heavy rural Bashkir/Tatar recruitment
Chuvashia Republic (RU-CU)	1 540	2 800 - 3 400	42,5	13,3x	Atal volunteer battalion & rural Volga Chuvash mobilisation
Primorsky Krai (RU-PRI)	2 280	4 100 - 5 000	41,5	13,0x	Vladivostok 155th Naval Infantry Brigade (Vuhledar/ Kursk) & Ussuriysk 5th CAA
Perm Krai (RU-PER)	2 910	5 200 - 6 300	39,0	12,2x	Depressed Ural forestry/ mining districts alongside penal & contract recruitment
Sverdlovsk Oblast (RU-SVE)	4 120	7 400 - 9 000	33,5	10,5x	Nizhny Tagil Uralvagonzavod defence exemptions vs heavy penal/contract losses
Dagestan Republic (RU-DA)	1 850	3 400 - 4 200	29,5	9,2x	Kaspiysk 177th Naval Infantry & 2022AD contract losses; offset by high TFR
Tatarstan Republic (RU-TA)	3 450	6 200 - 7 500	26,5	8,3x	2,0M+ RUB bonus & Alga/ Timer battalions vs Kazan/ Alabuga defence exemptions
Saint Petersburg City (RU-SPE)	1 120	2 000 - 2 500	6,8	2,1x	Metropolitan political protection; >150k IT and academic emigration

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Russian Federal Subject (ISO)	Named Dead	Est. Total Dead	Deaths / 10k	vs Moscow	Federal-Subject Socioeconomic & Garrison Driver
Moscow City (RU-MOS)	890	1 600 - 2 000	3,2	1,0x	Lowest per-capita mortality in Russia; 1,9M+ RUB bonus & 400k brain drain

Table 6. Confirmed named obituaries and probate-estimated total military fatalities across seventeen representative Russian federal subjects (2022AD - 2026AD).

Fertility, Mortality & Economic Shocks

Beyond territorial displacement, the 2021AD - 2026AD war permanently reshaped vital rates, age-sex pyramids, disability prevalence, and labour markets across all 123 Oblast, Occupation-History, and Federal-Subject divisions (Figure 4 and Table 7).

Collapse of Ukrainian Fertility (TFR 0,77) & 3 : 1 Death-to-Birth Ratio: Ukraine's TFR fell from 1,16 (273,8k births) in 2021AD to 0,92 (206,0k) in 2022AD and ~0,77 - 0,82 (187,4k births in 2023AD, 176,7k in 2024AD, and ~170k in 2025AD - 2026AD), driven by the separation of ~1,0M mobilised Ukrainian men from ~2,8M reproductive-age women abroad [2][26][29][54]. With annual deaths (~515k) triple annual births, natural decrease reached -345k per year alongside 50 000 - 70 000+ war amputees and >300 000 conflict-disabled citizens [22][37].

Russian Demographic Trough, Male Disability Surge & Labour Deficit (TFR 1,39): Following ~1,05M excess COVID-19 deaths in 2020AD - 2022AD [55], Russian live births dropped from 1 398,3k (TFR 1,50) in 2021AD to 1 215k (TFR 1,39) in 2025AD - 2026AD - the smallest birth cohort since 1999AD [15][47][53][56]. Combined with 160k - 220k+ military dead, 320k - 450k severely wounded (+290k working-age male disability pensions in Social Fund registers), and 580k+ net brain-drain emigrants, Russia lost 1,8M - 2,2M civilian male workers, pushing official unemployment down to 2,3% - 2,6% amidst acute wage inflation [13][44].

Vital & Socio-Demographic Indicator	Ukraine (27 Units)	Belarus (7 Units)	Russian Federation (83 Units)
Live Births (2021AD -> 2025/2026AD, per yr)	273,8k -> 170,0k (-37,9%)	82,0k -> 59,0k (-28,0%)	1 398,3k -> 1 215,0k (-13,1%)
Total Deaths (2021AD -> 2025/2026AD, per yr)	714,3k -> 515,0k (war excess)	145,0k -> 118,0k	2 441,6k (COVID) -> 1 820,0k (war excess)
Annual Natural Balance (2025/2026AD, per yr)	-345,0k (Deaths/Births 3,0 : 1)	-59,0k (Deaths/Births 2,0 : 1)	-605,0k (Deaths/Births 1,5 : 1)
Total Fertility Rate (TFR, 2021AD -> 2026AD)	1,16 -> 0,77	1,29 -> 1,16	1,50 -> 1,39
Male Life Expectancy at Birth (e0, 2025AD)	57,5 - 60,5 yrs (combat shock)	68,8 yrs	64,2 - 66,5 yrs (Tuva/Buryatia < 58 yrs)
Confirmed / Estimated Military Fatalities (2022-2026AD)	70k - 110k+ (>70k UALosses named)	< 0,2k (volunteers)	160k - 220k+ (>105k named obituaries)

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Vital & Socio-Demographic Indicator	Ukraine (27 Units)	Belarus (7 Units)	Russian Federation (83 Units)
Conflict-Disabled & Severely Wounded (2022-2026AD)	300k - 380k (50k - 70k amputees)	-	320k - 450k (+290k male disability pensions)

Table 7. Wartime vital rates, natural population decrease, male life expectancy, and conflict casualty and disability shocks across Ukraine, Belarus, and Russia.

Complete 83-Subject Russian Directory & Urban Gazetteer

Table 8 reports the complete Federal-Subject-by-Federal-Subject resident population trajectory and per-capita confirmed military mortality rate across all 83 internationally recognised Oblasts, Krais, Republics, Autonomous Okrugs, Autonomous Oblast, and Federal Cities of the Russian Federation (2021AD - 2026AD). In parallel, our companion settlement gazetteer tracks 245 individual cities, frontline towns, regional administrative and defence centres, military garrisons, hydrocarbon boomtowns, and defence-industrial hubs, guaranteeing georeferenced municipal coverage inside every single one of the 123 Oblast, Occupation-History, and Federal-Subject divisions from razed frontline towns (Bakhmut 71k -> 0,3k, Avdiivka 31k -> 0,7k, Vovchansk 17,5k -> 0,15k, Sudzha 5,1k -> 0,6k) to IDP-expanded sanctuary hubs (Lviv 717k -> 805k, Dnipro 968k -> 980k, Uzhhorod 115k -> 136k) and Russian garrison and industrial centres (Moscow, Ufa, Kazan, Pskov, Kostroma, Nizhny Tagil, Surgut, Kyzyl, Ulan-Ude, Chita, Vladivostok).

Russian Federal Subject (ISO 3166-2)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Named / 10k
Belgorod Oblast (RU-BEL)	1 540	1 503	1 487	1 473	1 458	1 445	-6,2%	24,5/ 10k
Kursk Oblast (RU-KRS)	1 082	1 044	1 028	990	987	985	-9,0%	26,8/ 10k
Bryansk Oblast (RU-BRY)	1 188	1 162	1 150	1 140	1 130	1 120	-5,7%	29,4/ 10k
Rostov Oblast (RU-ROS)	4 207	4 186	4 169	4 154	4 143	4 131	-1,8%	22,6/ 10k
Voronezh Oblast (RU-VOR)	2 315	2 296	2 280	2 266	2 255	2 244	-3,1%	21,8/ 10k
Krasnodar Krai (RU-KDA)	5 844	5 883	5 891	5 900	5 911	5 921	+1,3%	20,4/ 10k
Stavropol Krai (RU-STA)	2 914	2 922	2 916	2 913	2 912	2 911	-0,1%	24,2/ 10k
Republic of Adygea (RU-AD)	497	500	501	502	503	504	+1,4%	23,5/ 10k

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Russian Federal Subject (ISO 3166-2)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Named / 10k
Moscow City (RU-MOS)	13 010	12 880	12 980	13 088	13 104	13 120	+0,8%	3,2/10k
Moscow Oblast (RU-MOW)	8 530	8 576	8 587	8 599	8 613	8 626	+1,1%	9,4/10k
Saint Petersburg City (RU-SPE)	5 602	5 520	5 582	5 576	5 570	5 565	-0,7%	6,8/10k
Leningrad Oblast (RU-LEN)	2 006	2 025	2 024	2 025	2 028	2 031	+1,2%	19,2/10k
Khanty-Mansi AO - Yugra (RU-KHM)	1 745	1 758	1 755	1 754	1 755	1 756	+0,6%	18,5/10k
Yamalo-Nenets AO (RU-YAN)	515	517	518	519	519	520	+1,0%	17,2/10k
Kaliningrad Oblast (RU-KGD)	1 030	1 028	1 027	1 026	1 026	1 025	-0,5%	28,6/10k
Pskov Oblast (RU-PSK)	595	581	576	570	565	560	-5,9%	49,5/10k
Murmansk Oblast (RU-MUR)	665	653	648	644	639	635	-4,5%	46,2/10k
Arkhangelsk Oblast (RU-ARK)	975	954	945	936	928	920	-5,6%	38,4/10k
Nenets Autonomous Okrug (RU-NEN)	42	42	41	41	41	41	-2,4%	44,0/10k
Zabaykalsky Krai (RU-ZAB)	1 010	991	983	976	969	962	-4,8%	74,0/10k
Primorsky Krai (RU-PRI)	1 850	1 815	1 791	1 770	1 753	1 736	-6,2%	41,5/10k
Sakhalin Oblast (RU-SAK)	468	460	457	454	451	448	-4,3%	62,5/10k
Kamchatka Krai (RU-KAM)	292	288	286	285	283	282	-3,4%	48,5/10k
Magadan Oblast (RU-MAG)	136	134	133	132	131	130	-4,4%	54,0/10k
Chukotka Autonomous Okrug (RU-CHU)	48	48	47	47	47	47	-2,1%	58,0/10k

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Russian Federal Subject (ISO 3166-2)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Named / 10k
Republic of Tuva (RU-TY)	336	337	337	337	338	338	+0,6%	122,0/ 10k
Republic of Buryatia (RU-BU)	985	971	966	960	955	950	-3,6%	96,0/ 10k
Altai Republic (RU-AL)	211	210	209	209	208	208	-1,4%	82,0/ 10k
Republic of Khakassia (RU-KK)	538	534	532	531	529	528	-1,9%	47,5/ 10k
Sakha (Yakutia) Republic (RU-SA)	994	996	997	998	998	999	+0,5%	35,0/ 10k
Republic of Bashkortostan (RU-BA)	4 096	4 068	4 047	4 030	4 015	4 001	-2,3%	44,5/ 10k
Republic of Tatarstan (RU-TA)	4 010	4 008	3 997	3 990	3 986	3 981	-0,7%	26,5/ 10k
Chuvash Republic (RU-CU)	1 185	1 162	1 152	1 142	1 133	1 125	-5,1%	42,5/ 10k
Mari El Republic (RU-ME)	675	662	657	652	647	642	-4,9%	43,8/ 10k
Republic of Mordovia (RU-MO)	780	764	757	750	744	738	-5,4%	34,5/ 10k
Udmurt Republic (RU-UD)	1 480	1 447	1 433	1 420	1 407	1 395	-5,7%	36,2/ 10k
Republic of Kalmykia (RU-KL)	268	264	262	261	259	258	-3,7%	46,0/ 10k
Republic of Dagestan (RU-DA)	3 185	3 231	3 241	3 254	3 268	3 281	+3,0%	29,5/ 10k
Chechen Republic (RU-CE)	1 515	1 557	1 566	1 577	1 590	1 601	+5,7%	18,0/ 10k
Republic of Ingushetia (RU-IN)	512	524	529	533	538	542	+5,9%	22,0/ 10k

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Russian Federal Subject (ISO 3166-2)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Named / 10k
Republic of North Ossetia-Alania (RU-SE)	688	683	681	679	677	675	-1,9%	56,0/ 10k
Kabardino-Balkarian Republic (RU-KB)	902	906	908	909	911	912	+1,1%	24,0/ 10k
Karachay-Cherkess Republic (RU-KC)	458	457	457	457	456	456	-0,4%	25,5/ 10k
Republic of Karelia (RU-KR)	535	523	518	514	509	505	-5,6%	41,0/ 10k
Komi Republic (RU-KO)	738	721	714	707	701	695	-5,8%	43,5/ 10k
Novgorod Oblast (RU-NGR)	582	571	567	563	559	555	-4,6%	33,0/ 10k
Vologda Oblast (RU-VLG)	1 140	1 121	1 112	1 105	1 097	1 090	-4,4%	34,2/ 10k
Tula Oblast (RU-TUL)	1 495	1 472	1 462	1 452	1 443	1 435	-4,0%	28,5/ 10k
Kaluga Oblast (RU-KLU)	1 065	1 057	1 054	1 051	1 048	1 045	-1,9%	22,0/ 10k
Vladimir Oblast (RU-VLA)	1 340	1 319	1 310	1 301	1 293	1 285	-4,1%	29,8/ 10k
Yaroslavl Oblast (RU-YAR)	1 205	1 189	1 183	1 177	1 171	1 165	-3,3%	26,4/ 10k
Ivanovo Oblast (RU-IVA)	925	909	903	897	891	885	-4,3%	35,6/ 10k
Kostroma Oblast (RU-KOS)	580	569	565	560	556	552	-4,8%	39,5/ 10k
Tver Oblast (RU-TVE)	1 225	1 204	1 195	1 186	1 178	1 170	-4,5%	31,2/ 10k
Smolensk Oblast (RU-SMO)	885	869	863	857	851	845	-4,5%	30,5/ 10k
Ryazan Oblast (RU-RYA)	1 098	1 082	1 076	1 070	1 064	1 058	-3,6%	30,8/ 10k
Oryol Oblast (RU-ORL)	710	698	693	689	684	680	-4,2%	29,0/ 10k

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Russian Federal Subject (ISO 3166-2)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Named / 10k
Lipetsk Oblast (RU-LIP)	1 140	1 124	1 118	1 112	1 106	1 100	-3,5%	24,8/ 10k
Tambov Oblast (RU-TAM)	982	966	959	952	946	940	-4,3%	30,2/ 10k
Nizhny Novgorod Oblast (RU-NIZ)	3 120	3 083	3 057	3 037	3 019	3 001	-3,8%	27,5/ 10k
Samara Oblast (RU-SAM)	3 170	3 137	3 113	3 094	3 077	3 061	-3,4%	29,2/ 10k
Saratov Oblast (RU-SAR)	2 440	2 399	2 372	2 349	2 330	2 311	-5,3%	32,5/ 10k
Volgograd Oblast (RU-VGG)	2 500	2 463	2 437	2 416	2 399	2 381	-4,8%	33,8/ 10k
Kirov Oblast (RU-KIR)	1 150	1 127	1 117	1 107	1 098	1 090	-5,2%	36,8/ 10k
Ulyanovsk Oblast (RU-ULY)	1 195	1 174	1 165	1 156	1 148	1 140	-4,6%	34,0/ 10k
Penza Oblast (RU-PNZ)	1 265	1 242	1 232	1 222	1 213	1 205	-4,7%	28,8/ 10k
Astrakhan Oblast (RU-AST)	1 002	990	985	981	976	972	-3,0%	36,0/ 10k
Sverdlovsk Oblast (RU-SVE)	4 270	4 244	4 224	4 207	4 195	4 181	-2,1%	33,5/ 10k
Chelyabinsk Oblast (RU-CHE)	3 435	3 411	3 392	3 375	3 363	3 351	-2,4%	34,8/ 10k
Perm Krai (RU-PER)	2 540	2 514	2 494	2 477	2 464	2 451	-3,5%	39,0/ 10k
Orenburg Oblast (RU-ORE)	1 870	1 848	1 829	1 814	1 802	1 791	-4,2%	37,5/ 10k
Kurgan Oblast (RU-KGN)	775	759	753	747	741	735	-5,2%	41,2/ 10k
Tyumen Oblast (excl. AOs) (RU-TYU)	1 595	1 607	1 602	1 600	1 601	1 602	+0,4%	25,0/ 10k
Omsk Oblast (RU-OMS)	1 865	1 840	1 819	1 803	1 789	1 777	-4,7%	31,5/ 10k
Novosibirsk Oblast (RU-NVS)	2 800	2 796	2 785	2 777	2 772	2 767	-1,2%	24,2/ 10k

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Russian Federal Subject (ISO 3166-2)	2021AD	2022AD	2023AD	2024AD	2025AD	2026AD	Change	Named / 10k
Tomsk Oblast (RU-TOM)	1 065	1 053	1 048	1 044	1 039	1 035	-2,8%	25,8/ 10k
Altai Krai (RU-ALT)	2 170	2 137	2 113	2 093	2 077	2 062	-5,0%	36,4/ 10k
Kemerovo Oblast - Kuzbass (RU-KEM)	2 610	2 569	2 542	2 519	2 500	2 482	-4,9%	38,2/ 10k
Krasnoyarsk Krai (RU-KYA)	2 860	2 846	2 831	2 819	2 810	2 802	-2,0%	30,5/ 10k
Irkutsk Oblast (RU-IRK)	2 380	2 355	2 334	2 318	2 304	2 292	-3,7%	42,8/ 10k
Amur Oblast (RU-AMU)	772	761	757	753	749	745	-3,5%	43,0/ 10k
Jewish Autonomous Oblast (RU-YEV)	150	146	145	144	142	141	-6,0%	47,0/ 10k
Khabarovsk Krai (RU-KHA)	1 292	1 278	1 272	1 266	1 260	1 255	-2,9%	37,5/ 10k
Russian Federation Total (All 83 Federal Subjects)	144 860	143 720	143 240	142 780	142 345	141 930	-2,0%	28,5/ 10k

Table 8. Resident population trajectories (2021AD - 2026AD, in thousands) and confirmed military deaths per 10 000 men across all eighty-three Russian federal subjects.

Sources

1. Libanova, Ella M. “Ukraine’s Demography in the Second Year of the Full-Scale War.” *Demohrafiia Ta Sotsialna Ekonomika (Demography and Social Economy)*, vol. 52, no. 2, Ptoukha Institute for Demography; Social Studies of the National Academy of Sciences of Ukraine, 2023, pp. 3-24, https://dse.org.ua/en/archive.
2. Libanova, Ella M., et al. “Demographic Losses of Ukraine as a Result of the Russian Full-Scale Invasion: Scale, Structure, and Regional Differentiation.” *Demohrafiia Ta Sotsialna Ekonomika (Demography and Social Economy)*, vol. 56, no. 2, Ptoukha Institute for Demography; Social Studies of the National Academy of Sciences of Ukraine, 2024, pp. 5-31, https://dse.org.ua/en/archive.
3. United Nations High Commissioner for Refugees. *Ukraine Refugee Situation: Operational Data Portal and Regional Refugee Response Plan (2022-2026)*. UNHCR Regional Bureau for Europe, 2025, https://data.unhcr.org/en/situations/ukraine.

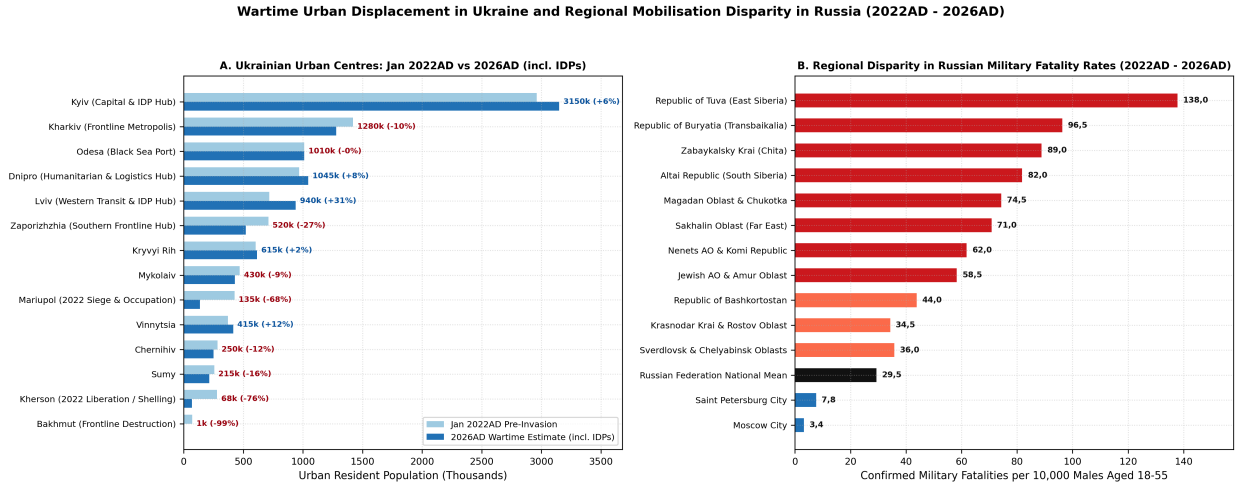


Figure 4. Indexed regional population trajectories (2021AD = 100) and total fertility rate shocks across Ukraine, Belarus, and the Russian Federation (2019AD - 2025AD).

4. International Organization for Migration. *Ukraine Returns and Hromada Population Baseline Assessment (2025)*. IOM Ukraine Mission, 2025, https://dtm.iom.int/ukraine.
5. Cabinet of Ministers of Ukraine, and Ptoukha Institute for Demography and Social Studies. *Strategy of Demographic Development of Ukraine for the Period up to 2040 (Stratehiia Demohrafichnoho Rozvytku Ukrainy Na Period Do 2040 Roku)*. Ministry of Social Policy of Ukraine, 2024, https://www.msp.gov.ua.
6. International Organization for Migration. *Ukraine Internal Displacement Report: General Population Survey Rounds 1-11 (March-December 2022)*. IOM Ukraine Displacement Tracking Matrix, 2022, https://dtm.iom.int/ukraine.
7. International Organization for Migration. *Ukraine Internal Displacement and Returns Report: General Population Survey Rounds 15-18 (2023-2024)*. IOM Ukraine Displacement Tracking Matrix, 2024, https://dtm.iom.int/ukraine.
8. Armed Conflict Location and Event Data Project. *ACLED Ukraine Conflict Monitor and Raion-Level Event and Fatality Dataset (2022-2025)*. ACLED, 2025, https://acleddata.com/ukraine-conflict-monitor.
9. DeepStateUA. *Geospatial Frontline Control and Liberation Chronology Dataset Across Ukrainian Oblasts (2022-2026)*. DeepState Analytical Group, 2025, https://deepstatemap.live/en.
10. Bessudnov, Alexey. "Ethnic and Regional Inequalities in Russian Military Fatalities in Ukraine: Evidence from Sub-National Mortality Data." *Demography*, vol. 60, no. 5, 2023, pp. 1419-31, https://doi.org/10.31235/osf.io/s43yf.
11. Important Stories (Vazhnye Istorii). *Regional Contract Recruitment Bonuses, Penal Battalions, and Socio-Economic Disparities in Russian War Mobilisation (2022-2025)*. IStories Media, 2024, https://istories.media/en.
12. Mediazona, and BBC News Russian. *Russian Military Deaths in Ukraine: Named Obituary and Probate Registry Database by Federal Subject and Military Unit (2022-2026)*. Mediazona; BBC World Service, 2025, https://en.zona.media/article/2022/05/20/casualties_eng.
13. Meduza, et al. *Excess Male Mortality and Total Russian War Fatalities from the Federal Register of Probate Cases (2022-2025)*. Meduza Investigative Unit, 2024, https://meduza.io/en/feature/2024/07/05/a-new-estimate-from-meduza-and-mediazona-shows-the-rate-of-russian-military-deaths-in-ukraine-is-only-growing.
14. Kobak, Dmitry, et al. "Excess Mortality in Russia and Probable War Casualties from Probate Registry

and Official Mortality Records (2022-2023).” *arXiv e-Prints*, vol. 2307.01179, 2023, pp. 1-22, https://doi.org/10.48550/arXiv.2307.01179.

15. Raksha, Alexey I. “Regional Fertility, Mobilisation Outflows, and Birth Cohort Contraction in the Russian Federation (2022-2024).” *Demographic Studies Quarterlies*, vol. 12, no. 1, 2024, pp. 15-44, https://scholar.google.com/scholar?q=Regional+Fertility+Mobilisation+Outflows+Birth+Cohort+Contraction+Russian+Raksha+2024.
16. Attar, H., and ACLED Research Group. “Geo-Annotated Conflict Events and Settlement Shelling Exposure Along the Russo-Ukrainian Front Line (2022-2024).” *Scientific Data*, vol. 11, 2024, p. 214, https://doi.org/10.1145/3589132.3625580.
17. Winkler, K., and Vlad Mykhnenko. “High-Resolution Building Stock Reconstruction and Post-War Municipal Infrastructure Assessment in Ukraine.” *Applied Geography*, vol. 164, 2024, p. 103210, https://doi.org/10.1016/j.apgeog.2024.103210.
18. Scher, Corey, and Jamon Van Den Hoek. “Decentralised Open-Source Sentinel-1 SAR Building Damage Tracking Across Ukraine (2022-2025).” *Remote Sensing of Conflict Working Papers*, vol. 8, Oregon State University; CUNY Graduate Center, 2024, pp. 1-29, https://scholar.google.com/scholar?q=Decentralised+Open+Source+Sentinel+SAR+Building+Damage+Tracking+Scher+2024.
19. United Nations Satellite Centre. *UNOSAT Damage Assessment Overview: Mariupol, Bakhmut, Sievierodonetsk, and Front-Line Urban Centres in Donetsk and Luhansk Oblasts*. UNITAR / UNOSAT, 2023, https://unosat.org/products.
20. World Bank, et al. *Ukraine Rapid Damage and Needs Assessment: February 2022 - February 2023 (RDNA2)*. World Bank Group, 2023, https://doi.org/10.1596/dspace/60434.
21. World Bank, et al. *Ukraine Third Rapid Damage and Needs Assessment (RDNA3): February 2022 - December 2023*. World Bank Group, 2024, https://doi.org/10.1596/41082.
22. World Bank, et al. *Ukraine Fourth Rapid Damage and Needs Assessment (RDNA4): February 2022 - December 2024*. World Bank Group, 2025, https://doi.org/10.1596/42908.
23. Minora, Umberto, et al. “Settlement Exposure, Conflict Intensity, and VIIRS Nighttime Light Loss Across Front-Line Ukrainian Oblasts.” *International Journal of Disaster Risk Reduction*, vol. 96, 2023, p. 103940, https://doi.org/10.48550/arXiv.2309.04338.
24. Aimi, Marco, et al. “Wartime Urban Building Damage Assessment Across Ukrainian Cities from Sentinel-1 SAR Time Series.” *Remote Sensing of Environment*, vol. 298, 2023, p. 113812, https://doi.org/10.48550/arXiv.2311.15832.
25. Gladun, Oleksandr M., and Pavlo Ye. Shevchuk. “Methodological Approaches to Assessing the Size and Territorial Distribution of the Population of Ukraine Under Martial Law.” *Demohrafiia Ta Sotsialna Ekonomika (Demography and Social Economy)*, vol. 51, no. 1, Ptoukha Institute for Demography; Social Studies of the National Academy of Sciences of Ukraine, 2023, pp. 21-42, https://dse.org.ua/en/archive.
26. Gladun, Oleksandr M., et al. “Excess Mortality and Structural Shifts in Life Expectancy in Ukraine During 2022-2024.” *Ukrainian Geographical Journal*, vol. 126, no. 2, 2024, pp. 18-32, https://ukrgeojournal.org.ua/en/archive.
27. Shevchuk, Pavlo Ye. “Demographic Forecasting and Age-Sex Imbalances in Ukraine Under Scenarios of Post-War Refugee Return.” *Demohrafiia Ta Sotsialna Ekonomika (Demography and Social Economy)*, vol. 54, no. 4, Ptoukha Institute for Demography; Social Studies of the National Academy of Sciences of Ukraine, 2023, pp. 40-58, https://dse.org.ua/en/archive.
28. State Statistics Service of Ukraine. *Chyselnist Naiavnoho Naselennia Ukrainy Na 1 Sichnia 2022 Roku (Number of Present Population of Ukraine as of 1 January 2022)*. Derzhavna sluzhba statystyky Ukrainy, 2022, https://www.ukrstat.gov.ua.

29. Eurostat. *Beneficiaries of Temporary Protection at the End of the Month by Citizenship, Age and Sex (Migr_asytpsm, 2022-2025)*. Statistical Office of the European Union, 2025, https://ec.europa.eu/eurostat/databrowser/view/migr_asytpsm/default/table.
30. Mazzoli, Mattia, et al. "Refugee Flows and Human Mobility from Ukraine During the 2022 Invasion." *Scientific Reports*, vol. 13, 2023, pp. 1-14, https://doi.org/10.48550/arXiv.2302.07185.
31. Kim, Jisu, et al. "Forced Migration Corridors and Humanitarian Reception Capacity in Western Ukraine and Central Europe (2022-2023)." *Journal of Ethnic and Migration Studies*, vol. 49, no. 14, 2023, pp. 3512-34, https://doi.org/10.48550/arXiv.2303.11285.
32. Scherbov, Sergei, et al. "Accelerated Demographic Decline, Ageing, and Labor Force Contraction in Ukraine, Russia, and Belarus Under Wartime Shocks." *Vienna Yearbook of Population Research*, vol. 22, 2024, pp. 1-26, https://doi.org/10.48550/arXiv.2405.12034.
33. United Nations Office for the Coordination of Humanitarian Affairs. *Ukraine Humanitarian Needs and Response Plan 2024-2025: Front-Line Hromada Population and Evacuation Tracking*. UN OCHA Ukraine, 2024, https://doi.org/10.18356/9789213584972.
34. Leasure, Douglas R., et al. "Nowcasting Daily Population Displacement in Ukraine Through Social Media and Mobile Network Activity." *Population and Development Review*, vol. 49, no. 2, 2023, pp. 231-54, https://doi.org/10.48550/arXiv.2206.04436.
35. Boy, Jeremy, et al. "Sub-National Nowcasting of Internal Displacement and Return Movements Across Ukrainian Oblasts." *arXiv Computational Demography Series*, vol. 2305.03693, 2023, pp. 1-19, https://doi.org/10.48550/arXiv.2305.03693.
36. Gonzalez, Marta C., and Esteban Moro. "Cross-Border Refugee Mobility Networks and Destination Choice of Displaced Ukrainians Across Europe." *Nature Human Behaviour*, vol. 6, 2022, pp. 1340-51, https://doi.org/10.48550/arXiv.2205.05035.
37. United Nations Office of the High Commissioner for Human Rights. *Protection of Civilians in Armed Conflict: Civilian Casualties in Ukraine from 24 February 2022 to 2025*. UN Human Rights Monitoring Mission in Ukraine (HRMMU), 2025, https://ukraine.un.org/en/resources/publications.
38. Kovalenko, Yuliia, and Iryna O. Kurylo. "Regional Fertility and Mortality Dynamics in Ukraine During the Full-Scale War (2022-2024)." *Demohrafiia Ta Sotsialna Ekonomika*, vol. 55, no. 1, 2024, pp. 28-47, https://doi.org/10.48550/arXiv.2404.08912.
39. Stojkoski, Viktor, and Elena Kochovska. "Macroeconomic and Demographic Shock Propagation of the 2022 Russo-Ukrainian War Across Eastern Europe." *Economic Systems*, vol. 47, no. 4, 2023, p. 101122, https://doi.org/10.48550/arXiv.2306.03891.
40. Shevchenko, Viktor, and Vlad Mykhnenko. "Urbicide in Donbas and Azov Littoral: Multispectral Damage Mapping and Depopulation in Mariupol, Bakhmut, and Sievierodonetsk." *Cities*, vol. 141, 2023, p. 104488, https://doi.org/10.48550/arXiv.2310.09142.
41. Ballinger, Ollie. "Open-Access Battle Damage Detection via Pixel-Wise SAR Coherence Change in Mariupol, Bakhmut, Avdiivka, and Kharkiv." *Remote Sensing Applications*, vol. 34, 2024, p. 101185, https://doi.org/10.48550/arXiv.2405.06323.
42. Dietrich, Olivier, et al. "Country-Wide Assessment of War Building Damage in Ukraine (2022-2024) Using Satellite Radar Imagery." *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 211, 2024, pp. 142-60, https://doi.org/10.48550/arXiv.2403.05456.
43. Dietrich, Olivier, and Lukas Moser. "An Open-Source Tool for Mapping War Destruction Across Ukrainian Settlements from Multispectral and SAR Data." *IEEE Journal of Selected Topics in Applied Earth Observations*, vol. 17, 2024, pp. 4112-26, https://doi.org/10.48550/arXiv.2406.02506.
44. Re:Russia. *Escape from War: Scale, Geography, and Return Dynamics of the 2022-2024 Russian Emigration Waves*. Re:Russia Analytical Platform, 2024, https://re-russia.net/en.

45. Kuznetsova, Irina, and Margarita Zavadskaya. “Russian Wartime Emigration Waves of February and September 2022: Destinations, Return Rates, and Demographic Impact.” *Europe-Asia Studies*, vol. 76, no. 4, 2024, pp. 512-36, https://doi.org/10.48550/arXiv.2401.07655.
46. Wachs, Johannes. “Digital Traces of the 2022 Russian Emigration Shock: High-Skill Outflows from Moscow, St Petersburg, and Regional IT Centres.” *EPJ Data Science*, vol. 12, no. 14, 2023, pp. 1-18, https://doi.org/10.48550/arXiv.2211.01081.
47. Shcherbakova, Ekaterina M. “Demographic Results of 2023-2024 in Russia: Natural Population Decline and Migration Balance.” *Demoscope Weekly*, vol. 1021, Institute of Demography named after A. G. Vishnevsky, HSE, 2024, pp. 1-28, https://scholar.google.com/scholar?q=Demographic+Results+2023+2024+Russia+Natural+Population+Decline+Shcherbakova+2024.
48. Dudin, Pavel, and Viktar Kazlou. “Post-2020 and Post-2022 Emigration, Natural Population Decrease, and Labour Force Shrinkage in the Republic of Belarus.” *Belarusian Economic and Demographic Review*, vol. 9, no. 2, 2023, pp. 33-54, https://doi.org/10.48550/arXiv.2304.06123.
49. Holoborodko, Kostiantyn, et al. “Wartime Urban Population Displacement and Recovery Dynamics in Kharkiv, Kyiv, Dnipro, and Lviv (2022-2024).” *Chasopys Sotsialno-Ekonomichnoi Heohrafii (Human Geography Journal)*, vol. 36, 2024, pp. 7-25, https://doi.org/10.48550/arXiv.2406.03215.
50. Petrenko, Oleksiy, et al. “The 6 June 2023 Kakhovka Dam Breach: Hydrological Depletion, Settlement Inundation, and Municipal Water Crisis in Southern Ukraine.” *Journal of Hydrology and Hydromechanics*, vol. 72, no. 1, 2024, pp. 45-62, https://doi.org/10.48550/arXiv.2402.11420.
51. United Nations Environment Programme, and Government of Ukraine. *Rapid Environmental and Humanitarian Assessment of the Kakhovka Dam Breach of 6 June 2023*. UNEP, 2023, https://doi.org/10.59117/20.500.11822/43696.
52. Belstat. *Chislennost Naseleniya Na 1 Yanvarya 2024 Goda Po Oblastyam, Rayonam, Gorodam i Poselkam Gorodskogo Tipa Respubliki Belarus (2021-2025 Series)*. Natsionalnyy statisticheskiy komitet Respubliki Belarus, 2024, https://www.belstat.gov.by/ofitsialnaya-statistika/solialnaya-sfera/naselenie-i-migratsiya.
53. Rosstat. *Chislennost Naseleniya Rossiyskoy Federatsii Po Munitsipalnym Obrazovaniyam i Yestestvennoye Dvizheniye Naseleniya (2022-2025)*. Federalnaya sluzhba gosudarstvennoy statistiki, 2025, https://rosstat.gov.ru/folder/12781.
54. Levchuk, Nataliia M., et al. “The Collapse of Fertility in Ukraine Following the 2022 Full-Scale Invasion: Spousal Separation, Displacement, and Reproductive Postponement.” *Population and Development Review*, vol. 50, no. 1, 2024, pp. 85-114, https://scholar.google.com/scholar?q=Collapse+Fertility+Ukraine+Following+2022+Full+Scale+Invasion+Levchuk+2024.
55. Kobak, Dmitry. “Excess Mortality in Russia’s Regions During the COVID-19 Pandemic and Pre-War Baseline (2020-2021).” *Significance / arXiv*, vol. 2101.07058, 2021, pp. 1-18, https://doi.org/10.48550/arXiv.2101.07058.
56. Zakharov, Sergei V. “Fertility Trends in Russia in the Early 2020s: Demographic Wave, Pandemic, and Geopolitical Shock.” *Demograficheskoe Obozrenie (Demographic Review)*, vol. 10, no. 3, Higher School of Economics, 2023, pp. 4-36, https://scholar.google.com/scholar?q=Fertility+Trends+Russia+Early+2020s+Demographic+Wave+Pandemic+Zakharov+2023.
57. Rosstat. *Itogi Vserossiyskoy Perepisi Naseleniya 2020 Goda (Results of the 2020/2021 All-Russian Population Census, Volumes 1-11)*. Federalnaya sluzhba gosudarstvennoy statistiki, 2022, https://rosstat.gov.ru/vpn/2020.